



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS & COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER II) – 2016/2017
CS105 – Object Oriented Programming Practical (1 Credit)

Answer **ALL** Questions

Time Allowed: **THREE** Hours

Instructions

Create a new project in your **H:** drive and save it as SXXXXX, where SXXXXX is your registration number.

You are requested to implement an application similar to **GoCab** taxi-service (rider-based functions only). The details of the application are given below.

GoCab is a technology platform that connects driver-partners and riders (i.e., users) via a mobile app.

In cities where **GoCab** operates, a ride can be requested using a rider app. When a nearby driver accepts a request, app displays an estimated time of arrival for the driver who is heading to the user's pickup location. Then, the app notifies the user when the driver is about to arrive.

The app also provides information about the driver whom the user will ride with, including first name, vehicle type, and license plate number. This information helps the user to connect with the driver at your pickup location.

Figure 1 shows the main flow of the entire process.

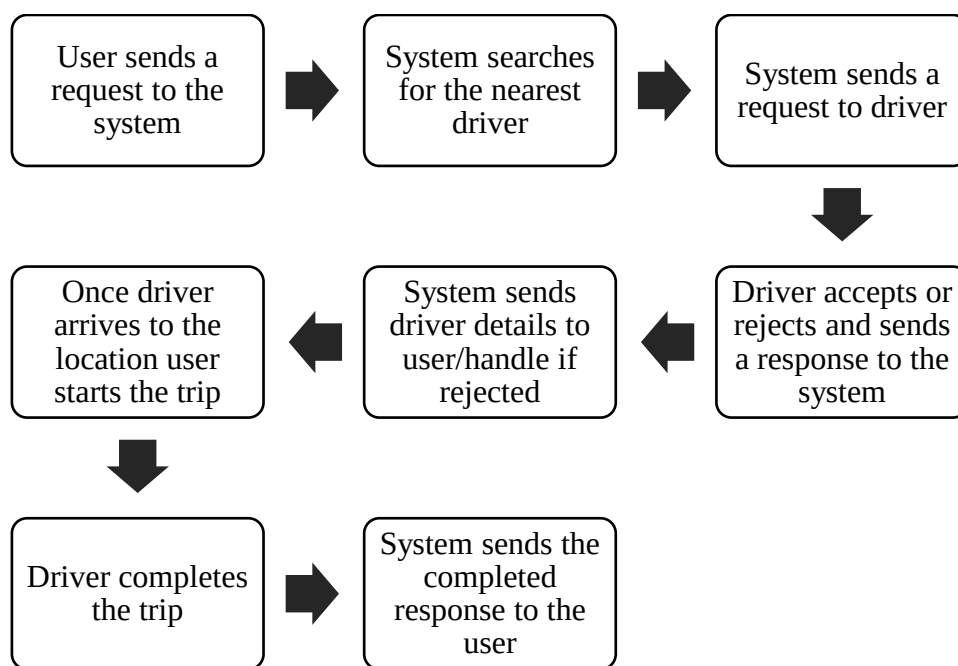


Figure 1 Main flow of the entire process

You have been given three sample text files, drivers.txt, user.txt and requests.txt. One sample row of each text file is shown below. Note that the actual text files contain comma separated data fields. Location's **latitude and longitude** are separated by “|” symbol

user.txt

User Id	User Name	Mobile	Rating	Completed Trips	Location
002	Kamal	0771222145	4.2	32	32.9697

					-96.80322
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driver.txt

User Id	User Name	Mobile	Rating	Trips	Location	Status	Car Type	Car Number	Car Model
003	Kamal	0771222145	4.2	32	32.9697 -96.80322	available	Hatchback	JD-5122	Toyota

request.txt

Request Id	Driver Id	User Id	Status
003	003	002	onride

You have to mainly implement the **functionalities of the user**. A sample project structure is shown in the **Figure 2**. You may use any object oriented concepts and best practices which are not mentioned here. **You may use the already implemented methods in your H:/ drive (SampleProgram) where necessary.**

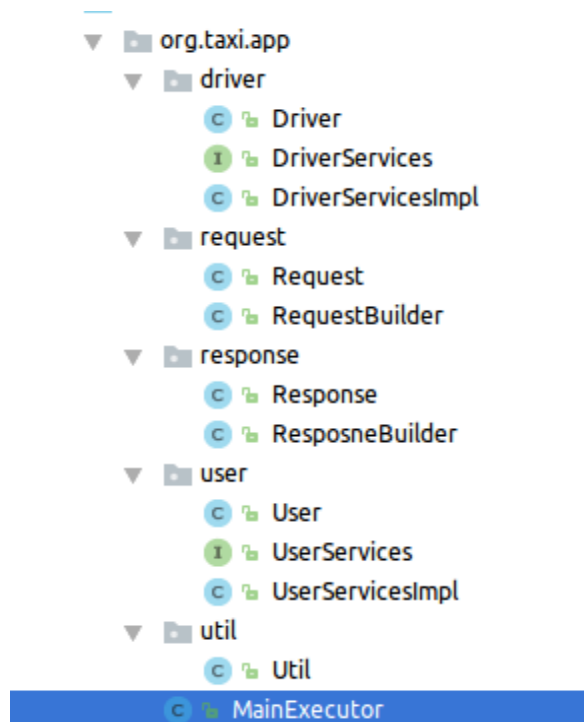


Figure 2 A sample project structure

[10 marks]

1. Read **users.txt** file and create all user objects and store them in a suitable data structure. [5 marks]
2. Read **drivers.txt** and create all driver objects and store them in a suitable data structure. [5 marks]
3. Read **requests.txt** and create all request objects in which the **state data field** is set to “**active**” and store them a in suitable data structure. [5 marks]
4. Implement a dashboard to perform all the user operations as illustrated in **Figure 3**. You may implement each of the steps given below. [10 marks]

Step 1:

Send a request to the system (including all the details of the user). There will be many requests coming into the system from different users. You have to store all the requests in a suitable data structure. **Update the request.txt file with all the data fields. [5 marks]**

Step 2:

Search for nearest drivers. You may implement a method to search the nearest driver using the dataset given in the drivers.txt. **[10 marks]**

Step 3:

Send nearest driver details to the user as a response object. **[5 marks]**

Step 4:

Start the timer for the ride. **[5 marks]**

Step 5:

Once user selects complete trip, you have to stop the timer and calculate the trip time and the cost. **[10 marks]**

Step 6:

Finally, user should be asked to provide a rating for the driver and **driver.txt** file should be updated with the calculated overall rating. **[10 marks]**

Expected Output Flow

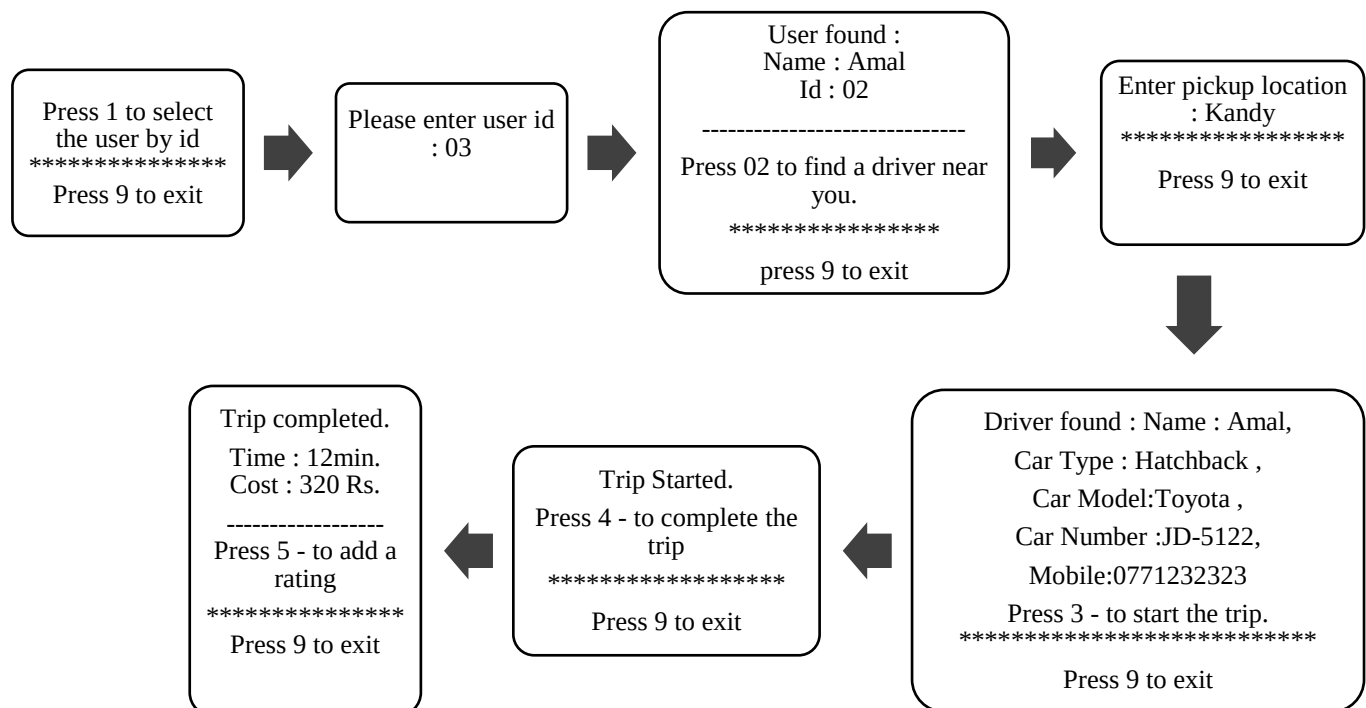


Figure 3 Expected dashboard output